Provision IaC and Deploy Web Application to AKS

The document describes a web application deployment to Azure Kubernetes (AKS) on Azure.

Tools and applications:

- Azure Cloud
- Azure DevOps
- ASP .NET Core Web Application
- Visual Studio Code
- Azure CLI
- Kubectl
- Terraform

Provision IaC using Terraform

The terraform will provision the services below:

- Azure Continuer Registry (ACR)
- Azure Kubernetes Cluster (AKS)
- Azure SQL Server
- Empty database

Create Service Principal with Contributor role in Azure AD:

az ad sp create-for-rbac --role="Contributor" --scopes="subscriptions/133bc1d5-767e-4628-a9d1-0914ff59665c" --"name="TerraformServicePrincipal"

Create backend storage in Azure for terraform.state:

TerraformStorage.ps1.txt

The powershell script creates the storage and provides **access key** like this:
==xaoKqIfd8j8y2txJCfat95rRMMKG15vKkZsX5PTuoKUn801d7mXo/qpkgPRPpPBlSLAPpHxAUTwcRFJK9NbmFg

Set the access key in the environment variable:

setx ARM_ACCESS_KEY
==xaoKqIfd8j8y2txJCfat95rRMMKG15vKkZsX5PTuoKUn801d7mXo/qpkgPRPpPBlSLAPpHxAUTwcRFJK9NbmFg

Create basic main.tf for ACR, AKS and SQL Server with empty Database:

main.tf.txt

terraform init

terraform plan

terraform apply

Check the AKS is up and running:

az aks get-credentials --resource-group rgyaronz --name aksyaronz --overwrite-existing

kubectl get nodes

Deploy the Web Application to AKS

Clone the web application source code from:

https://github.com/yaronzlot/EmployeeManagement

git clone https://github.com/yaronzlot/EmployeeManagement.git

Update the Azure SQL Server connection string in the web app code and update the empty database in Azure (using EF migration):

dotnet ef database update

dotnet build

dotnet run

http://localhost:5000 – verify the web application works against the update database in Azure SQL Server

Create Dockerfile:

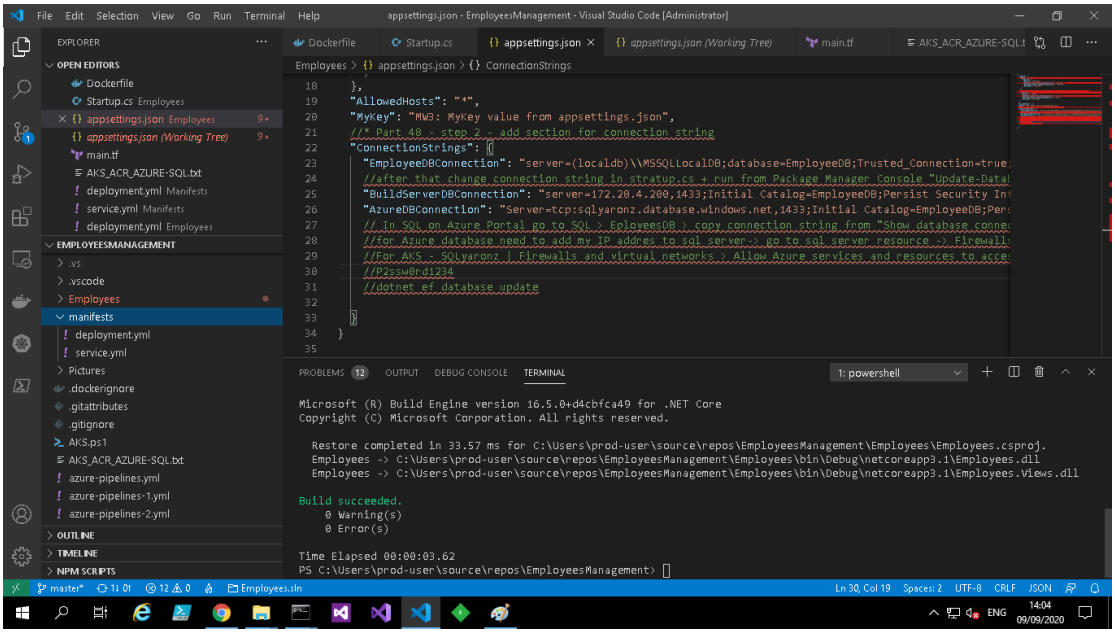
Dockerfile.txt

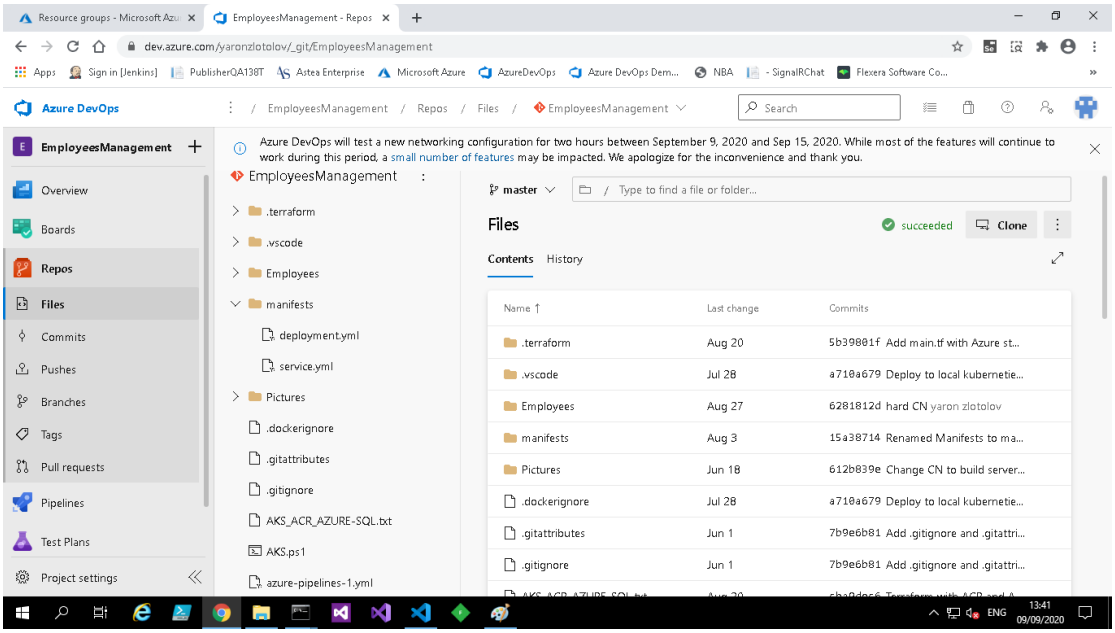
Create deployment.yml and service.yml in manifest folder:

deployment.yml.txt

service.yml.txt

Create project in Azure DevOps and push the code into the Azure GIT repo:





Create the pipeline (below is the pipeline yml):

EmployeesManagement.yml

NOTE: Before this step it is required to remove locks and IP restrictions from AKS done by Terraform:

az lock list

az lock delete –ids $lockid

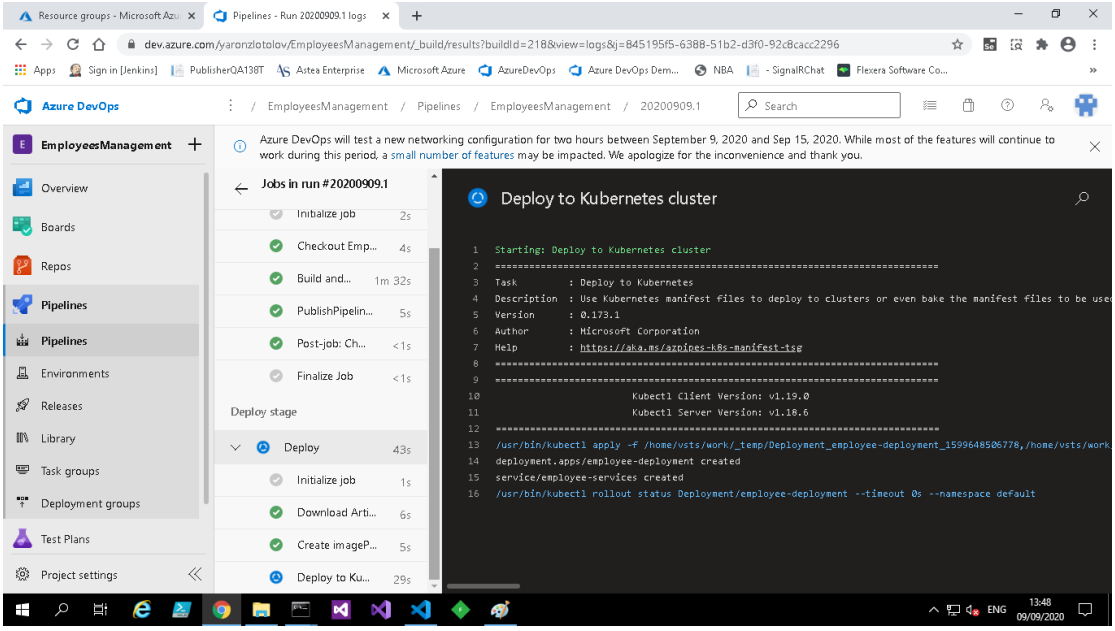
az aks update --resource-group rgyaronz --name aksyaronz --api-server-authorized-ip-ranges=

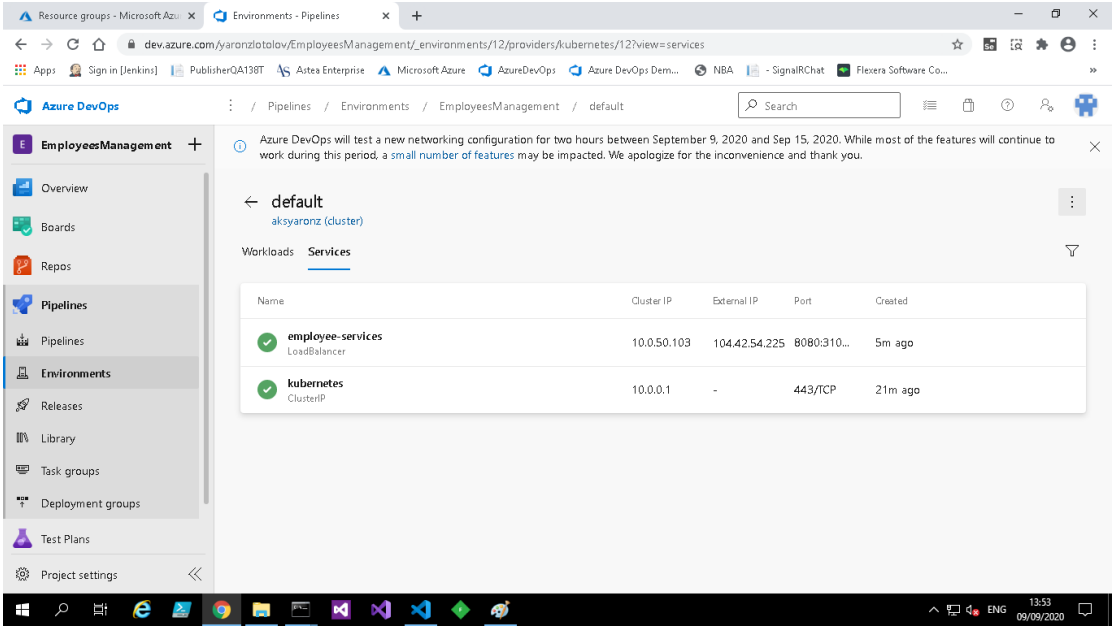
Build:

- Checkout the code from GIT repo
- Build the web application
- Create docker image
- Push the docker image to ACR
- Publish artifact for deployment

Deploy:

- Download the artifact
- Pull the image from ACR
- Deploy the docker image to AKS

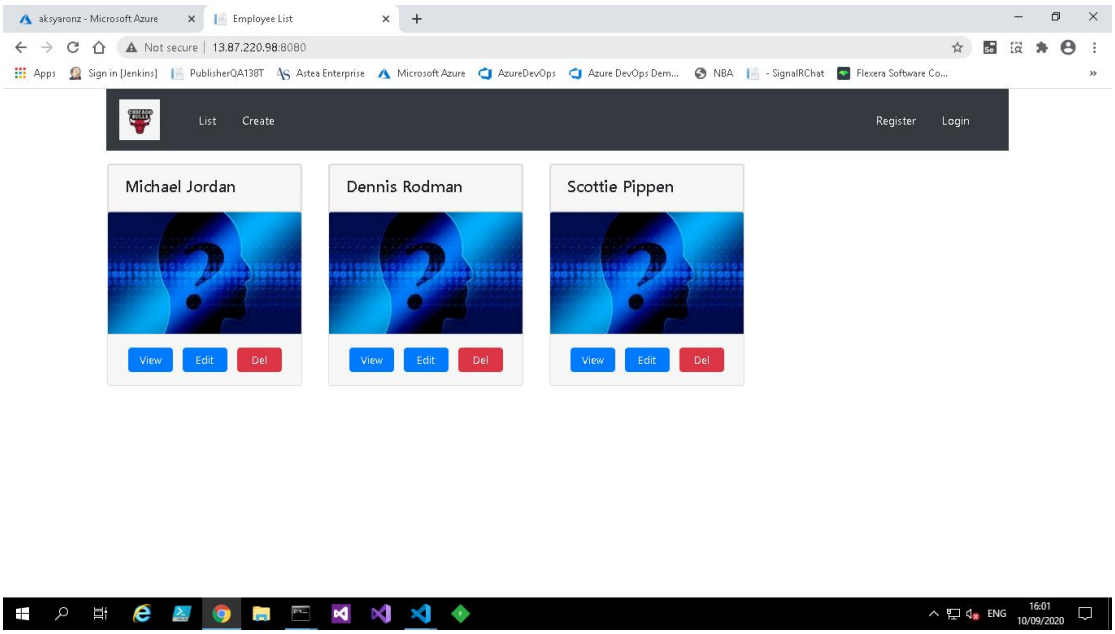




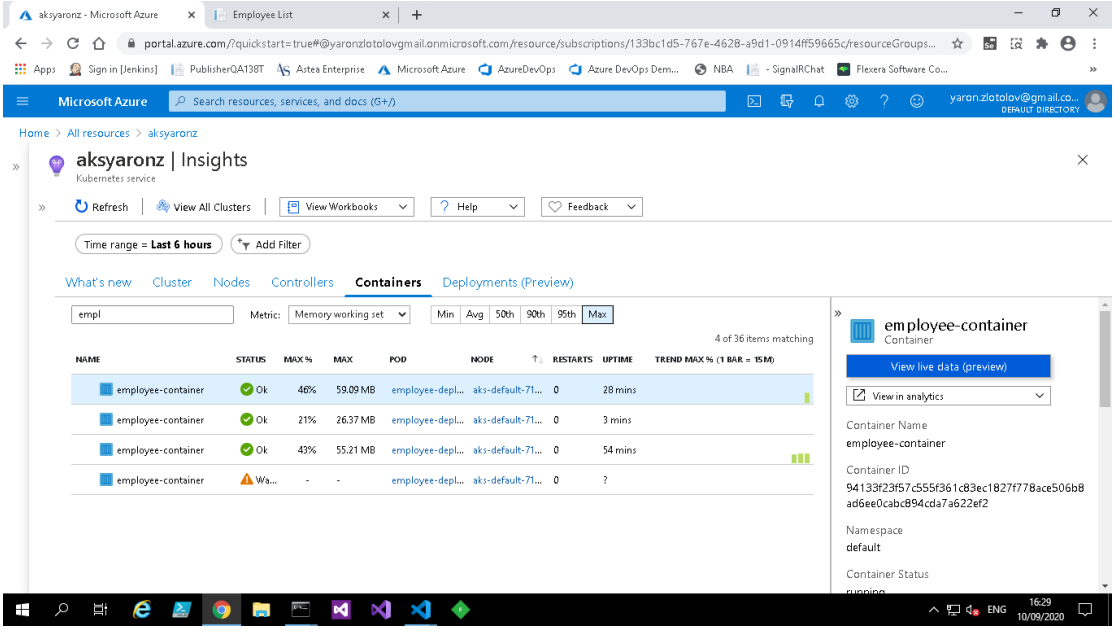
In case of problems with Azure DevOps need to deploy manually:

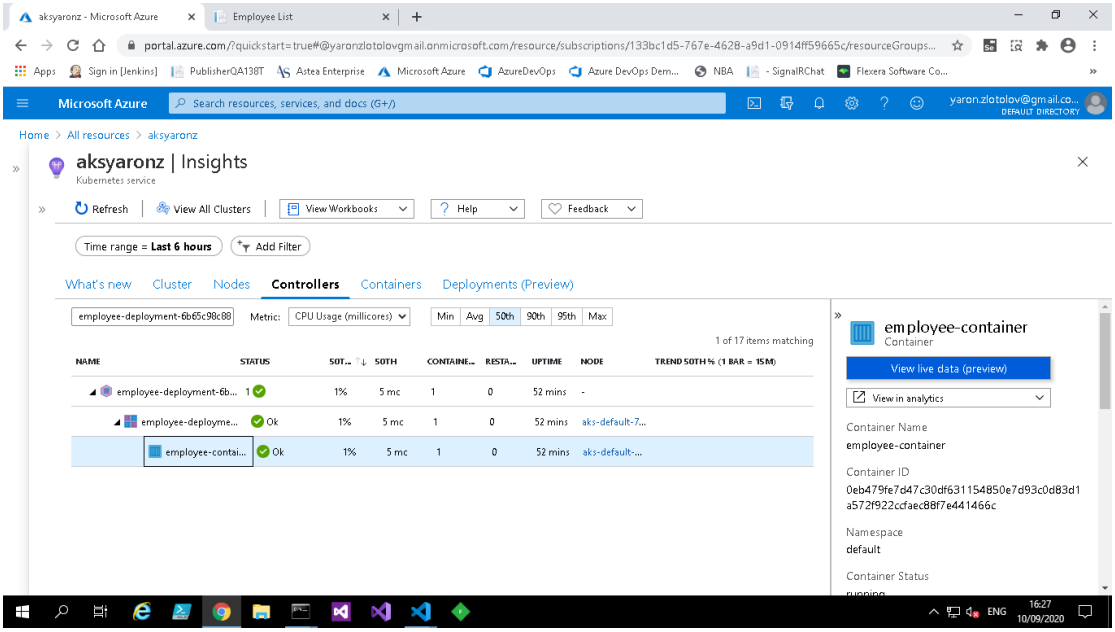
DeployToAKS.bat.txt

Web application is running on Azure:



Monitoring AKS Memory and CPU usage:





Done - Set the lock and the IP restriction again to AKS